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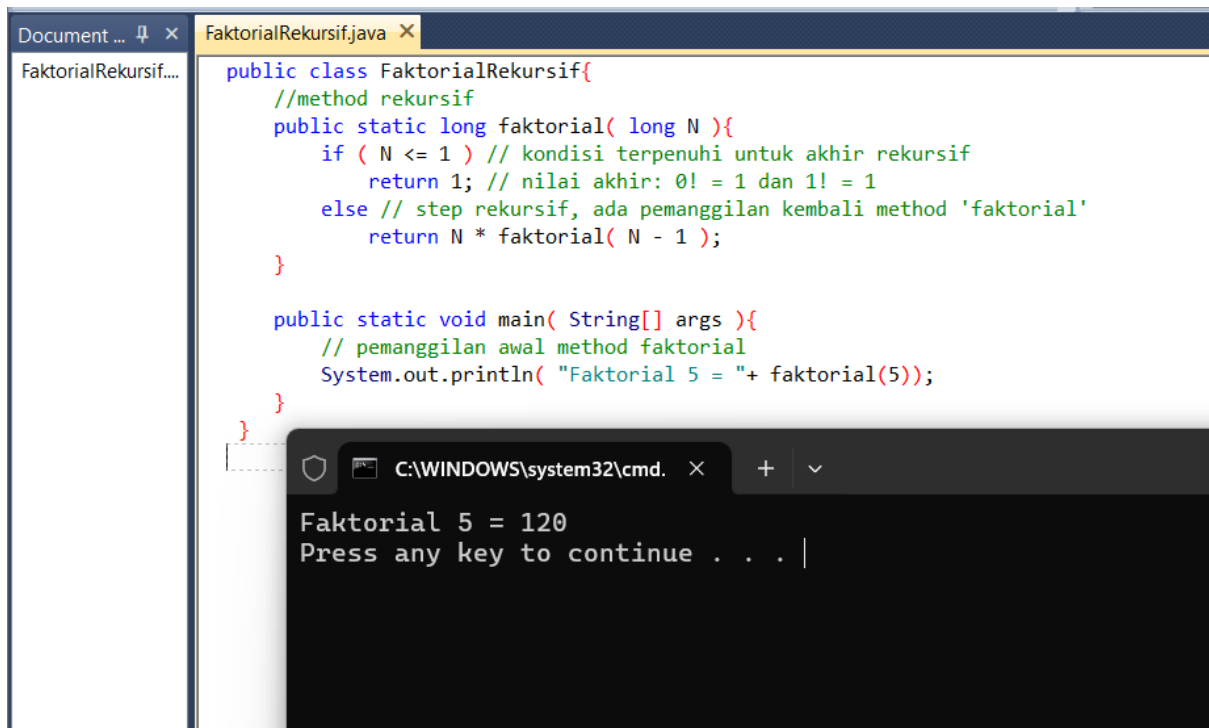
LISTING PRAKTIKUM ALGORITMA DAN PEMROGRAMAN LANJUT

MODUL 11

REKURSIF : definisi, cara kerja

PRAKTIK

Praktik 1



The screenshot displays a Java IDE with a file named `FaktorialRekursif.java`. The code defines a recursive method `faktorial` and a `main` method that calls it with the argument 5. Below the code, a command prompt window shows the output of the program.

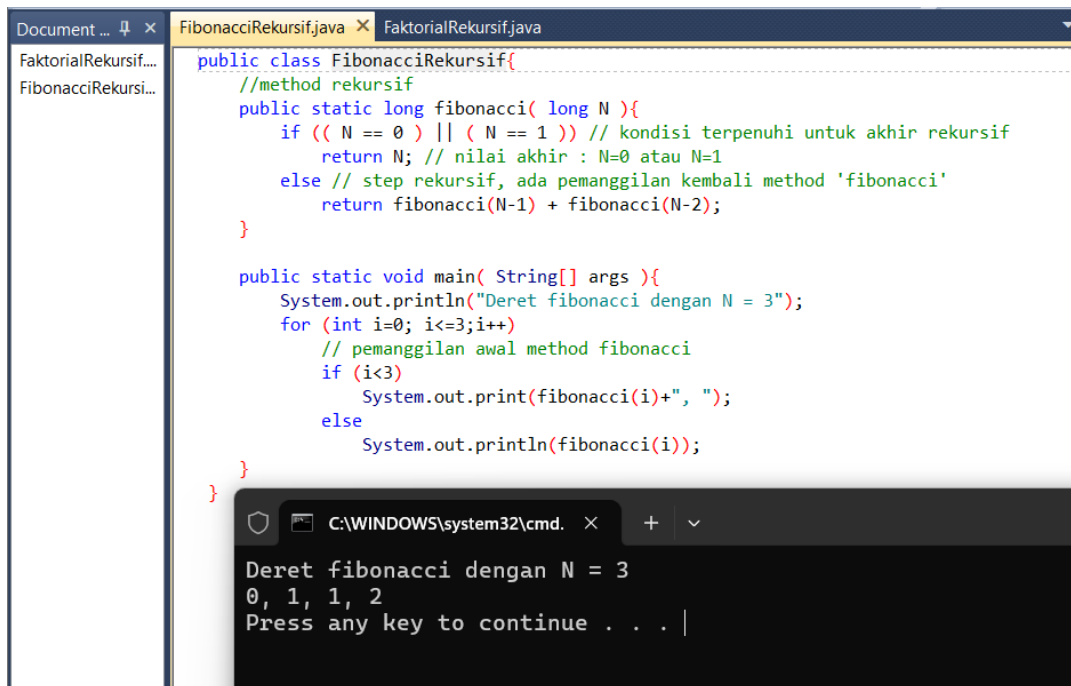
```
public class FaktorialRekursif{
    //method rekursif
    public static long faktorial( long N ){
        if ( N <= 1 ) // kondisi terpenuhi untuk akhir rekursif
            return 1; // nilai akhir: 0! = 1 dan 1! = 1
        else // step rekursif, ada pemanggilan kembali method 'faktorial'
            return N * faktorial( N - 1 );
    }

    public static void main( String[] args ){
        // pemanggilan awal method faktorial
        System.out.println( "Faktorial 5 = "+ faktorial(5));
    }
}
```

C:\WINDOWS\system32\cmd. x + v

Faktorial 5 = 120
Press any key to continue . . . |

Praktik 2



The screenshot shows an IDE with two tabs: `FibonacciRekursif.java` and `FaktorialRekursif.java`. The `FibonacciRekursif.java` tab is active, displaying the following Java code:

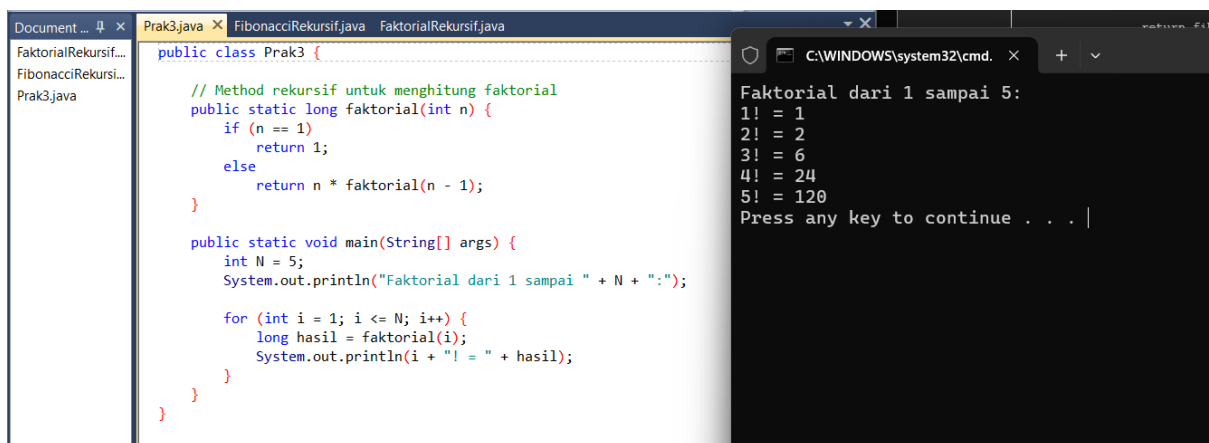
```
public class FibonacciRekursif{
    //method rekursif
    public static long fibonacci( long N ){
        if (( N == 0 ) || ( N == 1 )) // kondisi terpenuhi untuk akhir rekursif
            return N; // nilai akhir : N=0 atau N=1
        else // step rekursif, ada pemanggilan kembali method 'fibonacci'
            return fibonacci(N-1) + fibonacci(N-2);
    }

    public static void main( String[] args ){
        System.out.println("Deret fibonacci dengan N = 3");
        for (int i=0; i<=3;i++){
            // pemanggilan awal method fibonacci
            if (i<3)
                System.out.print(fibonacci(i)+", ");
            else
                System.out.println(fibonacci(i));
        }
    }
}
```

Below the code editor, a terminal window shows the output of the program:

```
Deret fibonacci dengan N = 3
0, 1, 1, 2
Press any key to continue . . . |
```

Praktik 3



The screenshot shows an IDE with three tabs: `Prak3.java`, `FibonacciRekursif.java`, and `FaktorialRekursif.java`. The `Prak3.java` tab is active, displaying the following Java code:

```
public class Prak3 {
    // Method rekursif untuk menghitung faktorial
    public static long faktorial(int n) {
        if (n == 1)
            return 1;
        else
            return n * faktorial(n - 1);
    }

    public static void main(String[] args) {
        int N = 5;
        System.out.println("Faktorial dari 1 sampai " + N + ":");

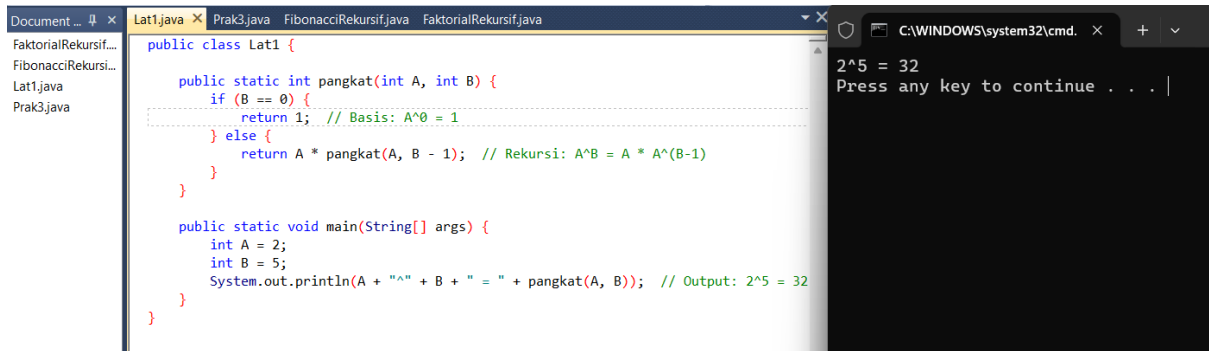
        for (int i = 1; i <= N; i++) {
            long hasil = faktorial(i);
            System.out.println(i + "! = " + hasil);
        }
    }
}
```

Below the code editor, a terminal window shows the output of the program:

```
Faktorial dari 1 sampai 5:
1! = 1
2! = 2
3! = 6
4! = 24
5! = 120
Press any key to continue . . . |
```

LATIHAN

Latihan 1



```
Document ... 4 x Lat1.java x Prak3.java FibonacciRekursif.java FaktorialRekursif.java
FaktorialRekursif...
FibonacciRekursi...
Lat1.java
Prak3.java

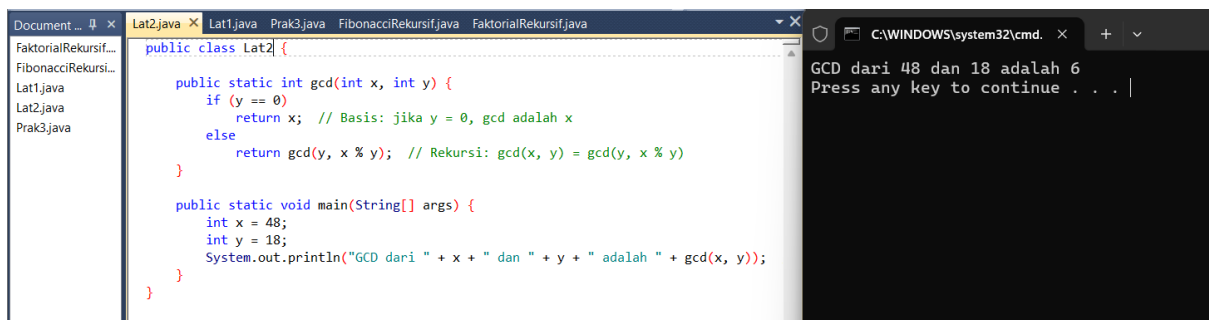
public class Lat1 {

    public static int pangkat(int A, int B) {
        if (B == 0) {
            return 1; // Basis: A^0 = 1
        } else {
            return A * pangkat(A, B - 1); // Rekursi: A^B = A * A^(B-1)
        }
    }

    public static void main(String[] args) {
        int A = 2;
        int B = 5;
        System.out.println(A + "^" + B + " = " + pangkat(A, B)); // Output: 2^5 = 32
    }
}
```

```
C:\WINDOWS\system32\cmd. x + v
2^5 = 32
Press any key to continue . . . |
```

Latihan 2



```
Document ... 4 x Lat2.java x Lat1.java Prak3.java FibonacciRekursif.java FaktorialRekursif.java
FaktorialRekursif...
FibonacciRekursi...
Lat1.java
Lat2.java
Prak3.java

public class Lat2 {

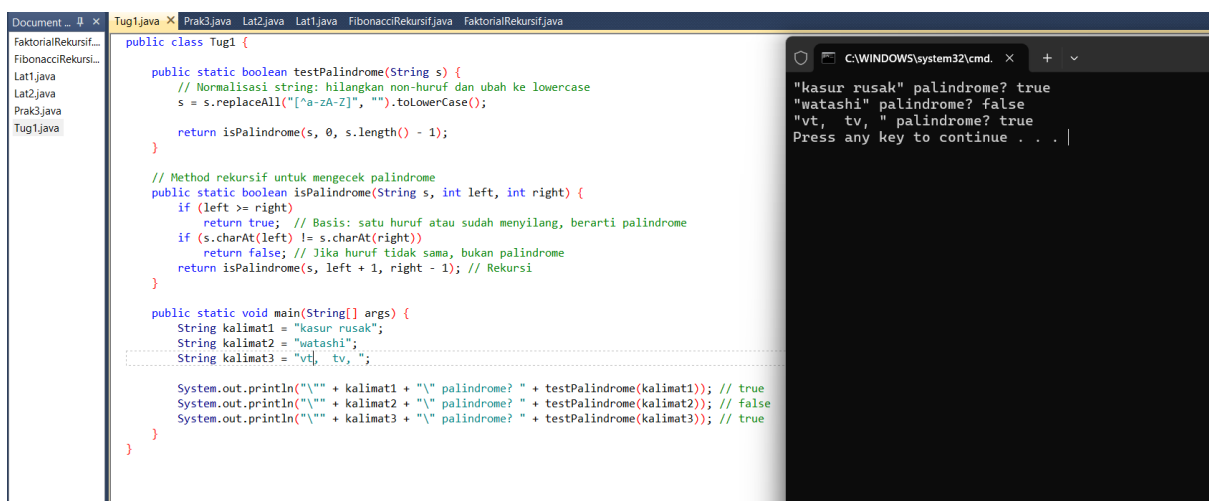
    public static int gcd(int x, int y) {
        if (y == 0) {
            return x; // Basis: jika y = 0, gcd adalah x
        } else {
            return gcd(y, x % y); // Rekursi: gcd(x, y) = gcd(y, x % y)
        }
    }

    public static void main(String[] args) {
        int x = 48;
        int y = 18;
        System.out.println("GCD dari " + x + " dan " + y + " adalah " + gcd(x, y));
    }
}
```

```
C:\WINDOWS\system32\cmd. x + v
GCD dari 48 dan 18 adalah 6
Press any key to continue . . . |
```

TUGAS

Tugas 1



```
Document ... 4 x Tug1.java x Prak3.java Lat2.java Lat1.java FibonacciRekursif.java FaktorialRekursif.java
FaktorialRekursif...
FibonacciRekursi...
Lat1.java
Lat2.java
Prak3.java
Tug1.java

public class Tug1 {

    public static boolean testPalindrome(String s) {
        // Normalisasi string: hilangkan non-huruf dan ubah ke lowercase
        s = s.replaceAll("[^a-zA-Z]", "").toLowerCase();

        return isPalindrome(s, 0, s.length() - 1);
    }

    // Method rekursif untuk mengecek palindrome
    public static boolean isPalindrome(String s, int left, int right) {
        if (left >= right) {
            return true; // Basis: satu huruf atau sudah menyilang, berarti palindrome
        } if (s.charAt(left) != s.charAt(right)) {
            return false; // Jika huruf tidak sama, bukan palindrome
        } return isPalindrome(s, left + 1, right - 1); // Rekursi
    }

    public static void main(String[] args) {
        String kalimat1 = "kasur rusak";
        String kalimat2 = "watashi";
        String kalimat3 = "vt, tv, ";

        System.out.println("'" + kalimat1 + "' palindrome? " + testPalindrome(kalimat1)); // true
        System.out.println("'" + kalimat2 + "' palindrome? " + testPalindrome(kalimat2)); // false
        System.out.println("'" + kalimat3 + "' palindrome? " + testPalindrome(kalimat3)); // true
    }
}
```

```
C:\WINDOWS\system32\cmd. x + v
"kasur rusak" palindrome? true
"watashi" palindrome? false
"vt, tv, " palindrome? true
Press any key to continue . . . |
```

Tugas 2